Case II

Under damped

$R^2C^2 < 4LC$

$A e^{s_1 t} + B e^{s_2 t}$

s_1, s_2

$$= -\frac{RC \pm \sqrt{(-ve)}}{2LC}$$

$$= -\frac{RC \pm j\sqrt{4LC - R^2C^2}}{2LC}$$

$$= -\frac{R}{2L} \pm j\sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}$$

ω, ω_d, $\left(\frac{1}{2s}\right)^2$

(A) $e^{-\frac{R}{2L}t}$ $e^{j\omega_d t}$

Sinusoid w/ decaying amp

(B) $e^{-\frac{R}{2L}t}$ $e^{j\omega_d t}$

Case III

Critically damped

$R^2C^2 = 4LC$

$$s_1 = s_2 = -\frac{R}{2L}$$

$$\mathcal{L}^{-1}\left(\frac{1}{(s-s_1)^2}\right)$$

$$\mathcal{L}^{-1}\left(\frac{1}{(s+a)^2}\right) = t e^{-at}$$

time Const τ_n

L, C, $V_c(0_-) = V$

R, L, C, $V_c(0_-) = V$

$$(s_1, s_2) = \frac{-RC \pm \sqrt{R^2C^2 - 4LC}}{2LC}$$

$$I(s) = \frac{Vc}{s^2LC + sRC + 1}$$

s_1, s_2 roots of $(s-s_1)(s-s_2)$

$i(t)$

$\mathcal{L}^{-1}$

$$\mathcal{L}^{-1}\left(\frac{A}{s-s_1} + \frac{B}{s-s_2}\right)$$

Case I

Overdamped

s_1, s_2 both real ($\Leftrightarrow$ real & true)

$R^2C^2 > 4LC$

$$s_1, s_2 = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

$$= A e^{s_1 t} + B e^{s_2 t}$$

$$\frac{1}{s+a} \overset{\mathcal{L}^{-1}}{\Rightarrow} e^{-at}$$

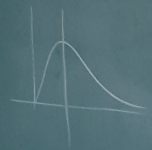
$e^{-\max(s_1, s_2)t}$

$RC > 2\sqrt{LC}$ ($\Rightarrow \frac{R}{L} > \frac{2}{\sqrt{LC}}$) $\Rightarrow \frac{1}{2L/R} > \frac{1}{\sqrt{LC}}$

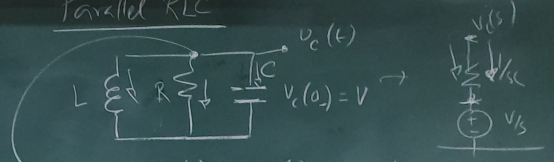
(P3) $\mathcal{L}^{-1}\left(\frac{1}{(s+a)^2}\right)$

$= \mathcal{L}^{-1}\left(\frac{1}{s+a}\cdot\frac{1}{s+a}\right)$

$= e^{-at}u(t) \otimes e^{-at}u(t)$



Parallel RLC



KCL : $\frac{V_c(s)}{R} + \frac{V_c(s)}{sL} + \frac{V_c(s) - V/s}{1/sC} = 0$

$\Rightarrow V_c(s)\left(\frac{1}{R} + \frac{1}{sL} + sC\right) = \frac{V}{s}\cdot\frac{1}{1/sC}$

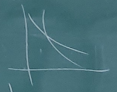
$\Rightarrow V_c(s)\left(s^2LCR + R + sL\right) = sLR\,Vc$

$\Rightarrow V_c(s) = \frac{s + \frac{1}{RC}V}{s^2 + \frac{1}{RC}s + \frac{1}{LC}}$

(P5) $\frac{A}{s-r_1} + \frac{B}{s-r_2}$ $\qquad r_1, r_2 = \frac{-\left(\frac{1}{RC}\right) \pm \sqrt{\left(\frac{1}{RC}\right)^2 - 4/LC}}{2}$

$= -\frac{1}{2RC} \pm \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}}$

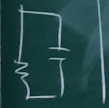
Case I (overdamped) $\left(\frac{1}{2RC}\right)^2 > \frac{1}{LC}$



Case II (underdamped) $\left(\frac{1}{2RC}\right) < \frac{1}{LC}$

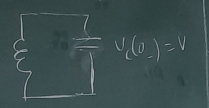


RC time Con



Case III (critically damped) $r = r$ $\qquad \mathcal{L}^{-1}\left(\frac{1}{(s-r_1)^2}\right) = te^{\mp\frac{1}{2RC}t}\left(e^{\mp\frac{1}{2RC}t}u(t) \otimes e^{\mp\frac{1}{2RC}t}u(t)\right)$

Last class $\frac{1}{\sqrt{LC}}$ was the ω in

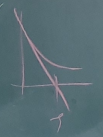


$A e^{\omega t} + B e^{-\omega t}$

$\omega = \frac{1}{\sqrt{LC}}$ frequency w/o resistor

– undamped freq

– natural freq



		Over-damped	Under-damped	Critically damped
Series RLC	$\zeta = \frac{1}{R}$	$\frac{1}{2\zeta} > \frac{1}{\sqrt{LC}} = \omega$	$\frac{1}{2\zeta} < \frac{1}{\sqrt{LC}}$ if ω	=
Parallel RLC	$\zeta = RC$	$\frac{1}{2\zeta} > \frac{1}{\sqrt{LC}} = \omega$ ⇒ $\zeta < \frac{1}{2\omega}$	$\frac{1}{2\zeta} < \frac{1}{\sqrt{LC}}$ if ω	=

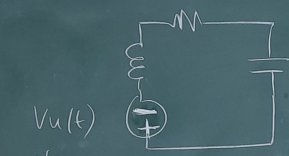
natural period $\zeta < \frac{\text{period}}{4\pi}$

$\frac{1}{\omega} = \frac{1}{2\pi}$ (T) period

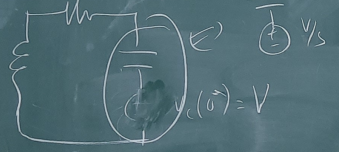
Oscillate

$\omega_d = \sqrt{\omega^2 - \frac{1}{(2\zeta)^2}}$

Series RLC



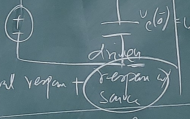
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Multiple sources



Superposition & study one at a time



add

last time

